

Moving on legs

Legs work like levers, thrusting an animal forward and keeping its body off the ground. The earliest known land animals with legs were millipedes, which have been on earth for over 400 million years. These pioneers were joined by six-legged insects, and then by four-legged vertebrates, which left water for a life on land. At first, they were slow-moving animals, but evolution produced longer, stronger legs, making predators better at chasing their food and helping prey to escape. Today, many hoofed mammals can run at more than 30 mph (50 kph), although that is sluggish compared to a sprinting cheetah.

Low slung

This spectacled salamander walks with its legs splayed out to the sides and its body slung close to the ground. Its body curls from side to side when it walks—a feature of movement that four-legged animals inherited from their fishy ancestors. Lizards also move in this sprawling way, but mammals have their legs under their bodies, which makes them the more efficient movers.

Splayed-out foot increases grip when walking up steep surfaces

Maximum legs

Millipedes have more legs than any other animal—up to an incredible 750. Their legs are tiny, and they have four on each body segment, unlike centipedes, which have two. When millipedes move, they push backward about 12 legs at a time in coordinated waves, which stops the legs from getting in each others' way.

Legs move in waves that start at the head

Killer sprint

The cheetah has a top speed of more than 68 mph (110 kph), thanks to its highly flexible spine, which arches up and down, extending its enormous stride. The cheetah hunts out in the open, accelerating from a standing start even faster than a car. However, a cheetah does not have great staying power. If the chase lasts more than a minute, it risks overheating, and its prey is usually able to escape.



Marathon mammals

Pronghorns are the fastest long-distance runners in the animal world. Their maximum speed is more than 53 mph (85 kph), but they can cruise across North America's open plains at 34 mph (55 kph) for more than 10 minutes without tiring.

Pronghorns are able to do this because they have an unusually large heart and lungs and small hooves, which minimize contact with the ground. Although they look more like antelopes, their closest living relatives are actually giraffes.

Armored exoskeleton shields legs when millipede coils up

